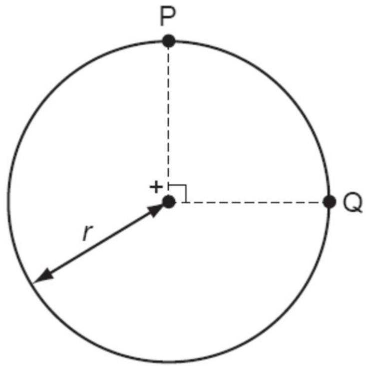
19 The diagram shows two points P and Q which lie 90° apart, on a circle of radius *r*.



A positive point charge at the centre of the circle creates an electric field of field strength *E* at both P and Q.

Which expression gives the work done in moving a unit positive charge from P to Q?

A 0

B $E \times r$

C $E \times \left(\dfrac{\pi r}{2}\right)$

D $E \times (\pi r)$

20 A lamp of resistance *R* is powered by an electrical source with internal resistance *r*.